

Report 2 hand-in

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Headings have been inserted in the document for readability. You only have to edit the part which says **(your solution here)** .

1 Master Yodas pendulum (yoda.py)

Problem 1: Formulate Yodas pendulum as a linear problem

$$A = \begin{bmatrix} 0 & 1 \\ -\frac{g}{L} & 0 \end{bmatrix} \quad (1)$$

$$B = \begin{bmatrix} 0 \\ \frac{1}{mL^2} \end{bmatrix} \quad (2)$$

(your solution here)

Problem 2: State at a later time

To solve the first part, we proceed by unfolding the recursion. In the absence of input ($\mathbf{B} = 0$, $\mathbf{u} = 0$), the system reduces to

$$\dot{\mathbf{x}} = A\mathbf{x}.$$

For Euler discretization we have:

$$\mathbf{x}_{k+1} = \tilde{A}_0 \mathbf{x}_k, \quad \text{where } \tilde{A}_0 = I + \Delta A$$

For $N = 1$:

$$\mathbf{x}_1 = \tilde{A}_0 \mathbf{x}_0$$

For $N = 2$:

$$\mathbf{x}_2 = \tilde{A}_0 \mathbf{x}_1 = \tilde{A}_0^2 \mathbf{x}_0$$

Continuing this process, we obtain:

$$\mathbf{x}_N = \tilde{A}_0^N \mathbf{x}_0$$

Similarly, for exponential integration:

$$\mathbf{x}_{k+1} = A_0 \mathbf{x}_k, \quad \text{where } A_0 = e^{A\Delta}$$

For $N = 1$:

$$\mathbf{x}_1 = A_0 \mathbf{x}_0$$

For $N = 2$:

$$\mathbf{x}_2 = A_0 \mathbf{x}_1 = A_0^2 \mathbf{x}_0$$

Thus, in general:

$$\mathbf{x}_N = A_0^N \mathbf{x}_0$$

As for the second part we get:

$$\tilde{A}_0 = \begin{bmatrix} 1 & \Delta \\ -\frac{g}{L}\Delta & 1 \end{bmatrix}, \quad A_0 = e^{A\Delta} \approx I + A\Delta + \frac{(A\Delta)^2}{2!} + \dots \quad (3)$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & \Delta \\ -\frac{g}{L}\Delta & 0 \end{bmatrix} + \frac{1}{2!} \begin{bmatrix} -\frac{g}{L}\Delta^2 & 0 \\ 0 & -\frac{g}{L}\Delta^2 \end{bmatrix} + \dots \quad (4)$$

(your solution here)

Problem 4: Eigenvalues and powers

Let

$$\tilde{A}_0 = Q\Sigma Q^{-1}, \quad \Sigma = \begin{bmatrix} \tilde{\lambda}_1 & 0 \\ 0 & \tilde{\lambda}_2 \end{bmatrix}.$$

Then

$$\tilde{M} = \tilde{A}_0^N = (Q\Sigma Q^{-1})^N.$$

For $N = 2$:

$$\tilde{A}_0^2 = (Q\Sigma Q^{-1})(Q\Sigma Q^{-1}) = Q\Sigma \underbrace{Q^{-1}Q}_I \Sigma Q^{-1} = Q\Sigma^2 Q^{-1}.$$

For $N = 3$:

$$\tilde{A}_0^3 = \tilde{A}_0^2 \tilde{A}_0 = (Q\Sigma^2 Q^{-1})(Q\Sigma Q^{-1}) = Q\Sigma^2 \underbrace{Q^{-1}Q}_I \Sigma Q^{-1} = Q\Sigma^3 Q^{-1}.$$

Continuing this process, we obtain

$$\tilde{A}_0^N = Q\Sigma^N Q^{-1}.$$

Since the eigenvectors are orthonormal, we have $Q^{-1} = Q^T$, which ensures that $Q^{-1}Q = I$.

Thus,

$$\Sigma^N = \begin{bmatrix} \tilde{\lambda}_1^N & 0 \\ 0 & \tilde{\lambda}_2^N \end{bmatrix},$$

and therefore the eigenvalues of $\tilde{M}$ are $\tilde{\lambda}_1^N$ and $\tilde{\lambda}_2^N$. **(your solution here)**

Problem 5: Analytical expression of Eigenvalues using Euler discretization

We compute the eigenvalues from the characteristic equation

$$\det(\tilde{A}_0 - \lambda I) = 0.$$

This gives

$$\det \begin{bmatrix} 1 - \lambda & \Delta \\ -\frac{g}{L}\Delta & 1 - \lambda \end{bmatrix} = (1 - \lambda)^2 + \frac{g}{L}\Delta^2 = 0.$$

Solving for λ yields

$$\lambda_{1,2} = 1 \mp i\Delta\sqrt{\frac{g}{L}}.$$

(your solution here)

Problem 6: Bound using Euler discretization

Using Euler discretization we get the upper bound:

$$\|\mathbf{x}_N\| = \|\tilde{A}_0^N \mathbf{x}_0\| \leq \|\tilde{A}_0^N\| = \|\tilde{A}_0\|^N = (\max |\lambda_i|)^N = \left(\sqrt{1 + \frac{g}{L}\Delta^2} \right)^N,$$

where we used that $\|\mathbf{x}_0\| = 1$ and the matrix norm definition

$$\|A\| = \max_{\|x\|=1} \|Ax\|.$$

(your solution here)

Problem 7: Matrix norm of Exponential discretization (harder)

Using exponential integration we get

$$\|\mathbf{x}_N\| = \|A_0^N \mathbf{x}_0\| \leq \|A_0^N\| = \|A_0\|^N,$$

where we used that $\|\mathbf{x}_0\| = 1$.

We now compute the eigenvalues of $A_0 = e^{\Delta A}$. First compute the eigenvalues of ΔA :

$$\Delta A = \begin{bmatrix} 0 & \Delta \\ -\frac{g}{L}\Delta & 0 \end{bmatrix}.$$

The characteristic equation is

$$\det(\Delta A - \lambda I) = \lambda^2 + \frac{g}{L}\Delta^2 = 0,$$

which gives

$$\lambda = \pm i\sqrt{\frac{g}{L}}\Delta.$$

We now use the eigendecomposition $\Delta A = Q\Sigma Q^{-1}$, where

$$\Sigma = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}.$$

Using the definition of the matrix exponential,

$$e^{\Delta A} = I + \Delta A + \frac{(\Delta A)^2}{2!} + \dots,$$

and the fact that $(\Delta A)^n = Q\Sigma^n Q^{-1}$, we obtain

$$e^{\Delta A} = Q \left(I + \Sigma + \frac{\Sigma^2}{2!} + \dots \right) Q^{-1} = Q e^{\Sigma} Q^{-1}.$$

Since Σ is diagonal, we have

$$e^{\Sigma} = \begin{bmatrix} e^{\lambda_1} & 0 \\ 0 & e^{\lambda_2} \end{bmatrix}.$$

Thus, the eigenvalues of $A_0 = e^{\Delta A}$ are

$$\lambda_{1,2} = e^{\pm i\sqrt{\frac{g}{L}}\Delta}.$$

Since $|e^{i\theta}| = 1$, it follows that

$$\|A_0\| = \max |\lambda_i| = 1.$$

Therefore,

$$\|\mathbf{x}_N\| \leq 1.$$

(your solution here)

2 R2D2 and control (r2d2.py)

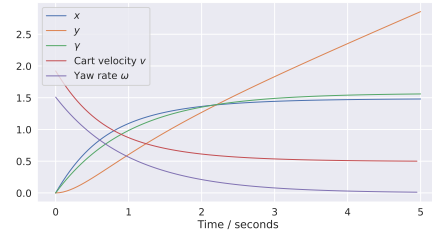
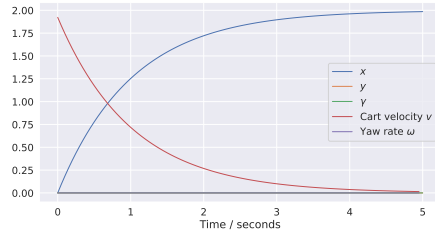
Problem 9: Discretization

$$\mathbf{x}_{k+1} = \mathbf{f}_k(\mathbf{x}_k, \mathbf{u}_k) = \begin{bmatrix} x_k + \Delta v_k \cos(\gamma_k) \\ y_k + \Delta v_k \sin(\gamma_k) \\ \gamma_k + \Delta \omega_k \end{bmatrix}$$

Problem 10: Linearization

$$\mathbf{x}_{k+1} \approx \begin{bmatrix} 1 & 0 & -\Delta \sin \theta \\ 0 & 1 & \Delta \cos \theta \\ 0 & 0 & 1 \end{bmatrix} \mathbf{x}_k + \begin{bmatrix} \Delta \cos \theta & 0 \\ \Delta \sin \theta & 0 \\ 0 & \Delta \end{bmatrix} \mathbf{u}_k + \begin{bmatrix} \Delta \theta \sin \theta \\ -\Delta \theta \cos \theta \\ 0 \end{bmatrix}$$

Problem 12: Control using simple linearization

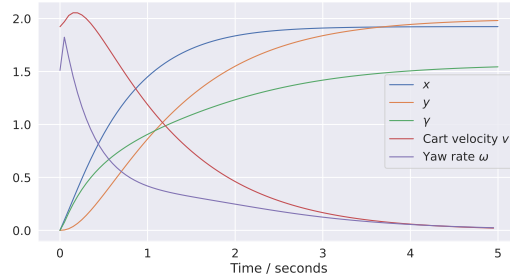


The system is linearized around the fixed point $\bar{x} = 0$, $\bar{u} = 0$.

For $x^* = [2, 0, 0]^T$, the motion remains close to $\gamma \approx 0$, where $\sin(\gamma) \approx 0$ and $\cos(\gamma) \approx 1$. Hence, the linear approximation is accurate and the controller performs well.

For $x^* = [2, 2, \frac{\pi}{2}]^T$, the system must reach $\gamma = \frac{\pi}{2}$, where $\sin(\gamma) = 1$ and $\cos(\gamma) = 0$. This is far from the linearization point, and the approximation $\sin(\gamma) \approx 0$, $\cos(\gamma) \approx 1$ becomes invalid. **(your solution here)**

Problem 13: MPC



In iterative linearization, the system is re-linearized at each time step around the current state and previous control. This means the linear model is always a local approximation of the true nonlinear dynamics.

As a result, the controller continuously adapts to the current state, capturing changes in orientation and position more accurately. In particular, the nonlinear effects of $\sin(\gamma)$ and $\cos(\gamma)$ are better approximated since the linearization point moves with the system. **(your solution here)**